



Spring Boot–REST Example

demo2-rest--boot: Files required

1. MyBootApplication.java	Same as Demo1
2. Hello.java	Same as Demo1
3. HelloController.java	Updated in Demo2
4. JLCAppConfig.java	Newly Added in Demo2
5. application.properties	Same as Demo1
6. pom.xml	Same as Demo1

3. HelloController.java

```
package com.jlccindia.springboot;
```

```
import org.springframework.web.bind.annotation.GetMapping;  
import org.springframework.web.bind.annotation.PathVariable;  
import org.springframework.web.bind.annotation.RequestParam;  
import org.springframework.web.bind.annotation.RestController;  
/*
```

```
* @Author : Srinivas Dande
```

```
* @company : Java Learning Center
```

```
*/
```

```
@RestController
```

```
public class HelloController {
```

```
    @GetMapping(value = "/myhello")
```

```
    public String getMessage1() {
```

```
        System.out.println("HelloController - getMessage1()");
```

```
        String msg = "Hello , I am getMessage1";
```

```
        return msg;
```

```
    }
```

```
    @GetMapping(value = "/myhello/{name}")
```

```
    public String getMessage2(@PathVariable String name) {
```

```
        System.out.println("HelloController - getMessage2()");
```

```
        String msg = "Hello " + name + " !!! - I am getMessage2";
```

```
        return msg;
```

```
    }
```

```
    @GetMapping(value = "/myhello/name/{name}", produces = { "application/json" })
```

```
    public Hello getMessage3(@PathVariable String name) {
```

```
        System.out.println("HelloController - getMessage3()");
```

```
        Hello hello = new Hello(101, name, "I am getMessage3");
```

```
        return hello;
```

```
    }
```



```
@GetMapping(value = "/myhello/name/{name}/message/{msg}", produces =
{"application/xml" })
public Hello getMessage4(@PathVariable String name, @PathVariable String msg) {
    System.out.println("HelloController - getMessage4()");

    Hello hello = new Hello(109, name, msg);
    return hello;
}

@GetMapping(value = "/myhello1/name/{name}", produces = { "application/xml",
    "application/json" })
public Hello getMessage5(@PathVariable String name) {
    System.out.println("HelloController - getMessage5()");

    Hello hello = new Hello(12345, name, "I am getMessage5");

    return hello;
}

@GetMapping(value = "/myhello1")
public Hello getMessage6(@RequestParam(required = false) String name) {
    System.out.println("HelloController - getMessage6()");

    Hello hello = new Hello(12345, name, "I am getMessage6");

    return hello;
}
}
```

3. JLCAppConfig.java

```
package com.jlcindia.springboot;

import java.util.List;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.http.converter.HttpMessageConverter;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class JLCAppConfig implements WebMvcConfigurer {
```



```
@Override
public void configureMessageConverters(List<HttpMessageConverter<?>> messageConverters) {

    System.out.println("-----Server Side Converters-----");
    for (HttpMessageConverter<?> myconverter : messageConverters) {
        System.out.println(myconverter);
    }

}
}
```

demo3-rest-server – REST API Runs here

demo3-rest-server: Files required

1. MyBootApplication.java	Same as Demo2
2. Hello.java	Same as Demo2
3. HelloController.java	Updated in Demo3
4. JLCAppConfig.java	Same as Demo2
5. application.properties	Same as Demo2
6. pom.xml	Same as Demo2

3. HelloController.java

```
package com.jlcindia.springboot;

import java.io.IOException;
import java.util.Collections;
import java.util.Enumuration;
import java.util.List;

import javax.servlet.http.HttpServletRequest;

import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
```



@RestController

```
public class HelloController {
```

@GetMapping(value="/myhello1")

```
public Hello getMessage1() throws IOException {  
System.out.println("HelloController - getMessage1() => GET");
```

```
    Hello hello =new Hello(101,"Srinivas Dande", "I am getMessage1" );  
    return hello;  
}
```

@GetMapping(value="/myhello2")

```
public ResponseEntity<Hello> getMessage2(HttpServletRequest request) throws IOException {  
System.out.println("HelloController - getMessage2() => GET");
```

//Accessing Request Headers

```
Enumeration<String> enms= request.getHeaderNames();  
List<String> headerNames = Collections.list(enms);
```

```
for(String headerName : headerNames) {  
    System.out.println(headerName+"\t"+ request.getHeader(headerName));  
}
```

//Preparing Response Body

```
    Hello respBody = new Hello(109,"Srinivas Dande", "I am getMessage2" );
```

//Preparing Response Headers

```
    HttpHeaders respHeaders = new HttpHeaders();  
    //respHeaders.setContentType(MediaType.APPLICATION_XML);  
    respHeaders.add("my-trainer", "Srinivas Dande");  
    respHeaders.set("my-course", "Spring MicroServices");  
    respHeaders.set("my-access-token", "JLC12345");
```

```
    ResponseEntity<Hello> respEntity =  
    new ResponseEntity<Hello>(respBody,respHeaders, HttpStatus.OK);
```

```
    return respEntity;  
}
```

@PostMapping(value="/myhello1")

```
public Hello getMessage1(@RequestBody Hello hello) throws IOException {  
System.out.println("HelloController - getMessage1() => POST");
```

```
    hello.setId(12345);  
    hello.setMessage("I am getMessage1() = POST");
```

```
    return hello;  
}
```



```
@PostMapping(value="/myhello2")
```

```
public ResponseEntity<Hello> getMessage2(@RequestBody Hello body,HttpServletRequest  
request) throws IOException {
```

```
System.out.println("HelloController - getMessage2() => POST");
```

```
//Accessing Request Headers
```

```
Enumeration<String> enms= request.getHeaderNames();
```

```
List<String> headerNames = Collections.list(enms);
```

```
for(String headerName : headerNames) {
```

```
System.out.println(headerName+"\t"+ request.getHeader(headerName));
```

```
}
```

```
//Preparing Response Body
```

```
body.setId(12345);
```

```
body.setMessage("I am getMessage1() = POST");
```

```
//Preparing Response Headers
```

```
HttpHeaders respHeaders = new HttpHeaders();
```

```
//respHeaders.setContentType(MediaType.APPLICATION_XML);
```

```
respHeaders.add("my-trainer", "Srinivas Dande");
```

```
respHeaders.set("my-course", "Spring MicroServices");
```

```
respHeaders.set("my-access-token", "JLC12345");
```

```
ResponseEntity<Hello> respEntity =
```

```
new ResponseEntity<Hello>(body,respHeaders, HttpStatus.OK);
```

```
return respEntity;
```

```
}
```

```
}
```

demo3-rest-client – Calling REST API

demo3-rest-client: Files required

1. Hello.java	Same as Demo3-Server
2. MyTest.java	New One
3. Test1.java	New One
4. Test2.java	New One
5. Test3.java	New One
6. Test4.java	New One
7. Test5.java	New One
8. Test6.java	New One



2. MyTest.java

```
package com.jlcindia.springboot;

import java.util.List;
import org.springframework.http.converter.HttpMessageConverter;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class MyTest {
    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();

        System.out.println("-----Client Side Converters-----");
        List<HttpMessageConverter<?>> messageConvertersList =
            restTemp.getMessageConverters();

        for (HttpMessageConverter<?> myconverter : messageConvertersList) {
            System.out.println(myconverter);
        }
    }
}
```

3. Test1.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test1 {
    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();
        String URL="http://localhost:12345/myhello1";

        Hello hello = restTemp.getForObject(URL, Hello.class);
        System.out.println(hello);

    }
}
```



4. Test2.java

```
package com.jlcindia.springboot;

import org.springframework.http.HttpHeaders;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test2 {
    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();
        String URL="http://localhost:12345/myhello1";

        ResponseEntity<Hello> respEntity = restTemp.getForEntity(URL, Hello.class);
        int statusCode = respEntity.getStatusCodeValue();

        System.out.println(statusCode);
        if(statusCode ==200) {
            //Success
            //Access Response Headers
            HttpHeaders respHeaders = respEntity.getHeaders();
            System.out.println(respHeaders);
            String contentType = respHeaders.getContentType().toString();
            System.out.println(contentType);

            //Access Response Body
            Hello hello = (Hello) respEntity.getBody();
            System.out.println(hello);

        }else {
            //Failure
            System.out.println(respEntity.getStatusCode());
        }

    }
}
```



5. Test3.java

```
package com.jlcindia.springboot;

import java.util.Collections;

import org.springframework.http.HttpEntity;
import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpMethod;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test3 {

    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();
        String URL="http://localhost:12345/myhello2";

        //Add Request Headers
        HttpHeaders reqHeaders = new HttpHeaders();
        //reqHeaders.setContentType(MediaType.APPLICATION_XML);
        //reqHeaders.setAccept(Collections.singletonList(MediaType.APPLICATION_XML));

        reqHeaders.setContentType(MediaType.APPLICATION_JSON);
        reqHeaders.setAccept(Collections.singletonList(MediaType.APPLICATION_JSON));

        reqHeaders.add("my-trainer", "Srinivas Dande");
        reqHeaders.set("my-course", "Spring MicroServices");
        reqHeaders.set("payment-status", "none");

        HttpEntity request = new HttpEntity(reqHeaders);

        ResponseEntity<Hello> respEntity = restTemp.exchange(URL,HttpMethod.GET, request,
        Hello.class);

        //Access Response Headers
        HttpHeaders respHeaders = respEntity.getHeaders();
        System.out.println(respHeaders);
        for( String headerName : respHeaders.keySet()) {
            System.out.println(headerName+"\t"+respHeaders.get(headerName));
        }
    }
}
```



//Access Response Body

```
Hello hello = (Hello) respEntity.getBody();  
System.out.println(hello);  
}  
  
}
```

6. Test4.java

```
package com.jlcindia.springboot;  
  
import org.springframework.web.client.RestTemplate;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */  
public class Test4 {  
    public static void main(String[] args) {  
  
        RestTemplate restTemp = new RestTemplate();  
        String URL="http://localhost:12345/myhello1";  
  
        Hello reqBody = new Hello(0,"Srinivas",null);  
  
        Hello hello = restTemp.postForObject(URL,reqBody, Hello.class);  
        System.out.println(hello);  
  
    }  
}
```

7. Test5.java

```
package com.jlcindia.springboot;  
  
import org.springframework.http.HttpHeaders;  
import org.springframework.http.ResponseEntity;  
import org.springframework.web.client.RestTemplate;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */  
public class Test5 {  
    public static void main(String[] args) {  
  
        RestTemplate restTemp = new RestTemplate();  
        String URL="http://localhost:12345/myhello1";
```



```
Hello reqBody = new Hello(0,"Srinivas",null);

ResponseEntity<Hello> respEntity = restTemp.postForEntity(URL, reqBody,Hello.class);
int statusCode = respEntity.getStatusCodeValue();

System.out.println(statusCode);
if(statusCode ==200) {
//Success
//Access Response Headers
HttpHeaders respHeaders = respEntity.getHeaders();
System.out.println(respHeaders);
String contentType = respHeaders.getContentType().toString();
System.out.println(contentType);

//Access Response Body
Hello hello = (Hello) respEntity.getBody();
System.out.println(hello);

}else {
//Failure
System.out.println(respEntity.getStatusCode());
}

}
}
```

8. Test6.java

```
package com.jlcindia.springboot;

import java.util.Collections;
import org.springframework.http.HttpEntity;
import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpMethod;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test6 {
public static void main(String[] args) {

RestTemplate restTemp = new RestTemplate();
String URL="http://localhost:12345/myhello2";
```



//Prepare Request Headers

```
HttpHeaders reqHeaders = new HttpHeaders();
reqHeaders.setContentType(MediaType.APPLICATION_XML);
reqHeaders.setAccept(Collections.singletonList(MediaType.APPLICATION_XML));

//reqHeaders.setContentType(MediaType.APPLICATION_JSON);
//reqHeaders.setAccept(Collections.singletonList(MediaType.APPLICATION_JSON));

reqHeaders.add("my-trainer", "Srinivas Dande");
reqHeaders.set("my-course", "Spring MicroServices");
reqHeaders.set("payment-status", "none");
```

//Prepare Request Body

```
Hello reqBody = new Hello(0,"Srinivas",null);

HttpEntity request = new HttpEntity(reqBody,reqHeaders);

ResponseEntity<Hello> respEntity =
restTemp.exchange(URL,HttpMethod.POST, request, Hello.class);
```

//Access Response Headers

```
HttpHeaders respHeaders = respEntity.getHeaders();
System.out.println(respHeaders);
for( String headerName : respHeaders.keySet()) {
System.out.println(headerName+"\t"+respHeaders.get(headerName));
}
```

//Access Response Body

```
Hello hello = (Hello) respEntity.getBody();
System.out.println(hello);
```

```
}
```

```
}
```